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Smart Triage for Monte Carlo: Optimising Dosimetry Workflows via Unsupervised Machine Learning

Content

Introduction
Accurate medical radiation dosimetry is critical for patient safety and treatment efficacy, with Monte Carlo (MC) simulation representing the gold standard for dose calculation accuracy. However, MC methods are computationally expensive and time-consuming, posing a significant infrastructure barrier for resource-constrained clinical environments. As computational methods, such as machine learning (ML) and Monte Carlo, increasingly contribute to many aspects of radiation medicine, there is an opportunity to integrate smart Quality Assurance (QA) filters into the dosimetry pipeline. This study proposes an unsupervised ML framework to perform "Computational Triage." The model automatically identifies highly anomalous radiotherapy treatment blueprints that require MC recalculation, allowing standard plans to be safely processed by faster, traditional analytical algorithms.

Methodology
To ensure scalability, this study bypassed computationally heavy 3D DICOM downloads by querying the cloud-native Imaging Data Commons (IDC) BigQuery database. We extracted 1D physical dosimetry metadata from a cohort of 324 unique Non-Small Cell Lung Cancer (NSCLC) patient profiles. Feature engineering was applied to calculate a novel 'Complexity Score' defined as the ratio of beam energy (KVP) to Slice Thickness. An unsupervised ML algorithm, the Isolation Forest, was selected to detect protocol deviations without requiring prior human labelling. The model was trained on scaled physics features (KVP, Exposure, Slice Thickness, and Complexity Score). To ensure clinical transparency and avoid "black-box" decision-making, model interpretability was mapped using SHapley Additive exPlanations (SHAP) and manifold latent space projection (UMAP).

Results and Discussion
Exploratory Data Analysis (EDA) of the 324 patients revealed a universally standardized slice thickness (3.0 mm) across the cohort. Consequently, the unsupervised ML algorithm autonomously pivoted to identify protocol deviations exclusively in beam energy and radiation exposure. The Isolation Forest successfully isolated 10 extreme protocol deviations, representing 3.1% of the total cohort. Explainable ML analysis using SHAP confirmed that extreme variations in Radiation Exposure (mAs) and KVP were the primary drivers triggering the MC triage alerts. The decision boundary established by the ML model proved that the algorithm learned actual dosimetric principles rather than statistical noise. All flagged anomalies mapped to a specific manufacturer (Siemens), demonstrating the model's ability to highlight hardware-specific protocol variations.

Conclusion
This study demonstrates the viability of an ML-driven "Computational Triage" model for modern QA workflows. By implementing this lightweight, explainable ML filter, clinical institutions can theoretically reduce their heavy computational burden by nearly 97%. This framework ensures that time-intensive MC simulations are reserved exclusively for highly anomalous cases where traditional algorithms pose a dosimetric risk. Ultimately, this transparent approach supports global health equity by maximizing existing computational resources in developing oncology centres.

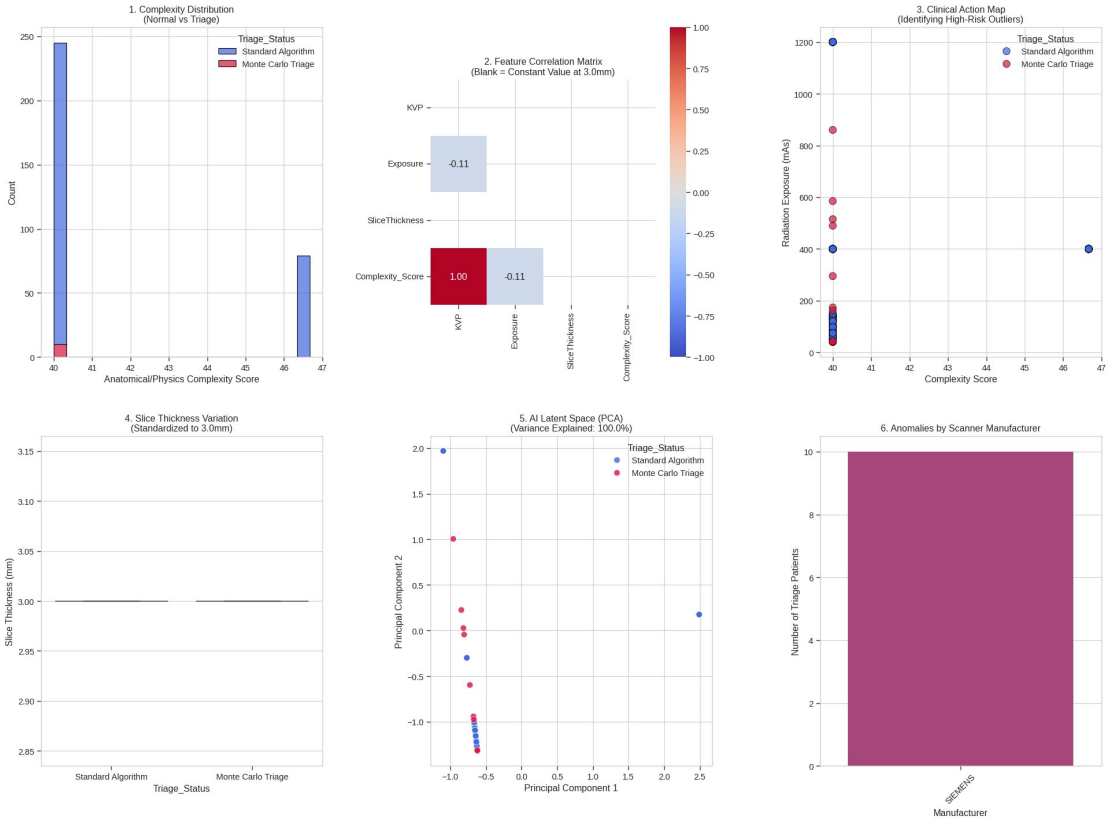


Figure 1: The 6-Panel EDA and Clinical Action Map showing the Geometric Decision Boundaries

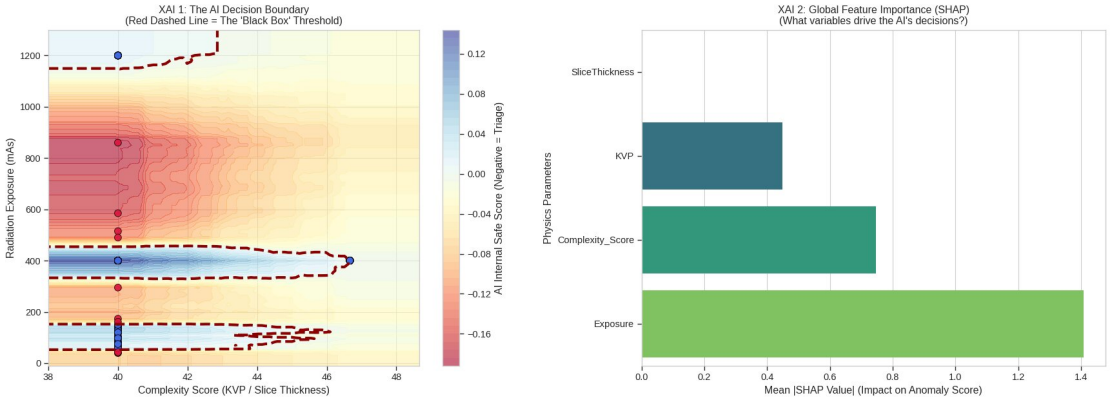


Figure 2: The SHAP Global Feature Importance Bar Chart

Alternative track

Country or Intl Organization

Malaysia

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